

Theorem H: the $s \rightarrow 0$ limit of Hikita's $\star$ is Hall–Littlewood e_k -multiplication

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Title: Theorem H: the $s \rightarrow 0$ limit of Hikita's $\star$ is Hall–Littlewood e_k -multiplication
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Grade: *proved* (self-proved; sober re-read once; not peer-reviewed). **Nov-
 elty UNAUDITED** — may be folklore (product-level shadow of $P_\lambda(x; q=0, t) = \text{HL}$).

1 Setup

Let $s = q^{-1}$ and $a_{ij} = (x_i - tx_j)/(x_i - x_j)$. The subset formula (Day 207b, proved and reviewed) is

$$E_k F = \sum_{|A|=k} \prod_A^\times X_A \cdot F(X_{A^c}, sX_A), \quad \prod_A^\times = \prod_{i \in A, j \notin A} a_{ij}, \quad (0.1)$$

where $E_k = e_k \star (-)$ and $X_A = \prod_{i \in A} x_i$. Put $e_\lambda^\star := E_{\lambda_1} \cdots E_{\lambda_\ell}(1)$ and $b_\mu := s^{n(\mu)} e_\mu$ (ordinary product e_μ). Here t is an indeterminate. By stability (Day 214 §5) we work in Λ_m with m at least the total degree.

Notation: $n(\kappa) = \sum (i-1)\kappa_i = \sum_{i < j} \min(\kappa_i, \kappa_j)$ over pairs of parts; for $\alpha \in \mathbb{Z}_{\geq 0}^m$, $\omega(\alpha) := \sum_j \binom{\alpha_j}{2}$, so $\omega(\alpha) = n(\kappa')$ for $\kappa = \text{sort } \alpha$; $\begin{bmatrix} N \\ r \end{bmatrix}_t$ is the Gaussian binomial.

Theorem H. Let $\mathcal{O} := \mathbb{Q}(t)[s]_{(s)}$ and $\mathcal{L} := \bigoplus_\mu \mathcal{O} b_\mu$.

(1) (Integrality) $E_k \mathcal{L} \subseteq \mathcal{L}$; more precisely $E_k b_\mu \in \bigoplus_\rho \mathbb{Q}[s, t] b_\rho$.

(2) (H) Let L_k be the $\mathbb{Q}(t)$ -linear map induced by E_k on $\mathcal{L}/s\mathcal{L} = \bigoplus \mathbb{Q}(t) \bar{b}_\mu$, and let $\varphi : \mathcal{L}/s\mathcal{L} \rightarrow \Lambda_{\mathbb{Q}(t)}$, $\varphi(\bar{b}_\mu) := t^{-n(\mu')} P_{\mu'}(x; t^{-1})$. Then

$$\varphi \circ L_k = t^{-\binom{k}{2}} e_k \cdot \varphi.$$

(3) (Explicit matrix) With $\rho := \mu'$, $\kappa := \nu'$,

$$L_k \bar{b}_\mu = \sum_{\nu} t^{\text{inv}(\kappa/\rho)} \prod_{v \geq 1} \begin{bmatrix} m_v(\kappa) \\ r_v \end{bmatrix}_t \bar{b}_\nu,$$

summed over $\kappa \supseteq \rho$ with κ/ρ a vertical k -strip; r_v is the number of parts of κ equal to v lying in rows of the strip, and $\text{inv}(\kappa/\rho) := \sum_{v < w} r_v (m_w(\kappa) - r_w)$. Equivalently ν/μ is a horizontal k -strip. The term $\nu = \mu \cup k$ (i.e. $\kappa = \rho + 1^k$) is always present; it is the DS leading term.

Corollary. For all $\lambda, \mu \vdash n$,

$$d_{\lambda\mu}(t) := [s^{n(\mu)}] c_{\lambda\mu} = t^{n(\mu') - n(\lambda')} [P_{\mu'}(x; t^{-1})] e_\lambda = t^{-n(\lambda')} \sum_{\nu} K_{\nu'\lambda} \tilde{K}_{\nu\mu'}(t),$$

where $e_\lambda^* = \sum_{\mu} c_{\lambda\mu} e_\mu$ and $\tilde{K}_{\nu\kappa}(t) = t^{n(\kappa)} K_{\nu\kappa}(t^{-1})$ is the cocharge Kostka–Foulkes polynomial. Hence $d_{\lambda\mu} \in \mathbb{N}[t]$; $d_{\lambda\mu}(1) = M_{\lambda\mu'}$, the number of 0-1 matrices with row sums λ and column sums μ' ; and $d_{\lambda\mu}(0) = 1$ for $\mu \supseteq \lambda$, recovering Day 214 Theorem 2(b).

2 Admissible polynomials are the b -lattice

Let $K = \mathbb{Q}(t)(s^{1/2})$ with s -adic valuation v (trivial on $\mathbb{Q}(t)$), extended to $K(x)$ by the Gauss valuation. For $c \in (\frac{1}{2}\mathbb{Z})^m$ let $\sigma_c : x_j \mapsto s^{-c_j} x_j$. A homogeneous symmetric $F = \sum_{\beta} a_{\beta} x^{\beta} \in K[x]$ is *admissible* if $v(a_{\beta}) \geq \omega(\beta)$ for all β .

Lemma 2.1. F is admissible $\iff$ for every c , $v(\sigma_c F) \geq \min_{\beta \in \mathbb{Z}_{\geq 0}^m} (\omega(\beta) - c \cdot \beta)$.

Proof. $(\Rightarrow)$ $v(\sigma_c F) = \min_{\beta} (v(a_{\beta}) - c \cdot \beta)$. $(\Leftarrow)$ Take $c = \alpha - \frac{1}{2}\mathbb{1}$. Then $\omega(\beta) - c \cdot \beta = \sum_j [(\beta_j - \alpha_j)^2 - \alpha_j^2]/2$, uniquely minimised on $\mathbb{Z}^m$ at $\beta = \alpha$. So $v(a_{\alpha}) - c \cdot \alpha \geq v(\sigma_c F) \geq \omega(\alpha) - c \cdot \alpha$. $\square$

Initial form. For admissible F put $\text{in}(F)(\alpha) := \text{residue of } s^{-\omega(\alpha)} a_{\alpha} \text{ in } \mathbb{Q}(t)$. It is symmetric in α , so a function of $\kappa = \text{sort } \alpha$, written $\text{in}(F)(\kappa)$. With $c = \alpha - \frac{1}{2}$ and $v_0 = \omega(\alpha) - c \cdot \alpha$, the proof above shows

$$\text{in}_{v_0}(\sigma_c F) = \text{in}(F)(\alpha) \cdot x^{\alpha} \quad \text{in the residue field } \mathbb{Q}(t)(x), \quad (1.2)$$

since every $\beta \neq \alpha$ contributes at strictly higher valuation.

Lemma 2.2. Let $F = \sum_{\mu} c_{\mu} e_{\mu} \in \Lambda_m \otimes K$ be homogeneous of degree $n \leq m$. Then F is admissible $\iff v(c_{\mu}) \geq n(\mu)$ for all μ , i.e. $F \in \mathcal{L} \otimes \mathcal{O}[s^{1/2}]$. In that case $\text{in}(F)(\mu') = \text{res}(s^{-n(\mu)} c_{\mu}) =: \bar{c}_{\mu}$, the $\bar{b}_{\mu}$ -coordinate of $F \bmod s$.

Proof. b_{μ} is admissible. The x^{β} -coefficient of $s^{n(\mu)} e_{\mu}$ is $s^{n(\mu)} M_{\mu\kappa}$, $\kappa = \text{sort } \beta$. By Day 214 Lemma 1.1, $M_{\mu\kappa} \neq 0 \Rightarrow \kappa \trianglelefteq \mu'$, so $\kappa' \trianglerighteq \mu$ (Macdonald I (1.11)), hence $\omega(\beta) = n(\kappa') \leq n(\mu)$, since n is strictly order-reversing ($n(\lambda) = \sum_{j \geq 1} (|\lambda| - S_j(\lambda))$). Equality iff $\kappa = \mu'$. So b_{μ} is admissible with $\text{in}(b_{\mu})(\kappa) = [\kappa = \mu']$, using $M_{\mu\mu'} = 1$.

$(\Leftarrow)$ Admissibility is preserved by sums and by scalars of valuation ≥ 0 .

$(\Rightarrow)$ We have $a_{\mu'} = \sum_{\nu \trianglelefteq \mu} c_{\nu} M_{\nu\mu'}$ with $M_{\mu\mu'} = 1$. Let $\delta := \min_{\nu} (v(c_{\nu}) - n(\nu))$ and suppose $\delta < 0$. Choose μ dominance-minimal with $v(c_{\mu}) - n(\mu) = \delta$. Every $\nu \triangleleft \mu$ has $v(c_{\nu}) \geq n(\nu) + \delta > n(\mu) + \delta$. Hence $v(a_{\mu'}) = n(\mu) + \delta < \omega(\mu')$, contradicting admissibility.

Formula. $\text{in}(F)(\mu') = \sum_{\nu} \text{res}(s^{n(\nu) - n(\mu)} \cdot s^{-n(\nu)} c_{\nu}) M_{\nu\mu'}$; only ν with $n(\nu) \leq n(\mu)$ and $\nu \trianglelefteq \mu$ survive, i.e. $\nu = \mu$. $\square$

3 The one-step recursion

Lemma 3.1 (level-set sum). *For $0 \leq r \leq N$,*

$$\sum_{B \subseteq [N], |B|=r} \prod_{i \in B, j \notin B} \frac{x_i - tx_j}{x_i - x_j} = \begin{bmatrix} N \\ r \end{bmatrix}_t.$$

Proof. The sum S is symmetric (permuting variables permutes terms), so $V_N S$ is an antisymmetric polynomial, divisible by V_N ; thus S is a polynomial, homogeneous of degree 0, hence a constant in $\mathbb{Q}(t)$. Substitute $x_i = u^{N-i}$ and let $u \rightarrow \infty$: $a_{ij} \rightarrow 1$ for $i < j$ and $\rightarrow t$ for $i > j$. So $S = \sum_B t^{\#\{i \in B, j \notin B: i > j\}}$, the inversion generating function of 0-1 words with r ones, which is $\begin{bmatrix} N \\ r \end{bmatrix}_t$. (Equivalently Macdonald III (1.4) split over $S_r \times S_{N-r}$ cosets.) $\square$

Proposition 3.2. *Let $F \in \Lambda_m \otimes K$ be admissible of degree n , $m \geq n + k$. Then $E_k F$ is admissible, and for every $\kappa \vdash n + k$*

$$\text{in}(E_k F)(\kappa) = \sum_{\rho: \kappa/\rho \text{ vert. } k\text{-strip}} t^{\text{inv}(\kappa/\rho)} \prod_v \begin{bmatrix} m_v(\kappa) \\ r_v \end{bmatrix}_t \text{in}(F)(\rho). \quad (2.3)$$

Proof. Fix c and A with $|A| = k$, and consider $\sigma_c R_A$ for the term R_A of (0.1).

1. *Kernel.* Since t is a unit, $v(\sigma_c(x_i - tx_j)) = v(\sigma_c(x_i - x_j)) = \min(-c_i, -c_j)$, so $v(\sigma_c a_{ij}) = 0$, with initial form 1 if $c_i > c_j$, t if $c_i < c_j$, and a_{ij} itself if $c_i = c_j$.
2. *Monomial.* $\sigma_c X_A = s^{-c \cdot \mathbb{K}_A} X_A$.
3. *Shifted F .* $\sigma_c[F(X_{A^c}, sX_A)] = \sigma_{c - \mathbb{K}_A} F$.
4. *Lower bound.* By Lemma 2.1 and $\omega(\beta + \mathbb{K}_A) = \omega(\beta) + \mathbb{K}_A \cdot \beta$,

$$v(\sigma_c R_A) \geq -c \cdot \mathbb{K}_A + \min_{\beta \geq 0} (\omega(\beta) - (c - \mathbb{K}_A) \cdot \beta) = \min_{\gamma \geq \mathbb{K}_A} (\omega(\gamma) - c \cdot \gamma) \geq \min_{\gamma \geq 0} (\omega(\gamma) - c \cdot \gamma).$$

Minimising over A , Lemma 2.1 gives admissibility of $E_k F$ (symmetric by 207b).

Now fix $\alpha \in \mathbb{Z}_{\geq 0}^m$ with $\text{sort } \alpha = \kappa$, $c = \alpha - \frac{1}{2}$, $v_0 = \omega(\alpha) - c \cdot \alpha$. All terms have $v \geq v_0$, so initial forms add and by (1.2) $\text{in}(E_k F)(\alpha) x^\alpha = \sum_A \text{in}_{v_0}(\sigma_c R_A)$.

5. $A \not\subseteq \text{supp } \alpha$. Then $\gamma = \alpha$ is excluded from $\{\gamma \geq \mathbb{K}_A\}$; the minimiser over $\mathbb{Z}^m$ is unique and the discrete minimum is attained, so $v(\sigma_c R_A) > v_0$: contribution 0.
6. $A \subseteq \text{supp } \alpha$. Put $\beta_0 = \alpha - \mathbb{K}_A \geq 0$, so $c - \mathbb{K}_A = \beta_0 - \frac{1}{2}$. By (1.2) for F at β_0 and multiplicativity, $\text{in}_{v_0}(\sigma_c R_A) = \text{in}_0(\sigma_c \prod_A^\times) \cdot \text{in}(F)(\beta_0) \cdot x^\alpha$; the levels match by the key identity.
7. *Kernel factorises.* Let $L_v = \{j : \alpha_j = v\}$, $m_v = |L_v|$, $B_v = A \cap L_v$, $r_v = |B_v|$ ($v \geq 1$). By step 1,

$$\text{in}_0(\sigma_c \prod_A^\times) = t^{\#\{i \in A, j \notin A: \alpha_i < \alpha_j\}} \prod_v \prod_{i \in B_v, j \in L_v \setminus B_v} a_{ij},$$

and the t -exponent is $\sum_{v < w} r_v(m_w - r_w) = \text{inv}$.

8. *Dependence only on (r_v) .* $\text{sort}(\beta_0) =: \rho(r)$ is κ with r_v of its parts equal to v lowered by one; $\kappa/\rho(r)$ is a vertical k -strip, and each vertical k -strip arises from a unique (r_v) . Both $\text{in}(F)(\beta_0) = \text{in}(F)(\rho(r))$ and inv depend only on (r_v) .
9. *Sum over the B_v .* For fixed (r_v) , Lemma 3.1 on each level set gives $\prod_v \begin{bmatrix} m_v \\ r_v \end{bmatrix}_t$. Dividing by x^α gives (2.3). $\square$

4 Proof of Theorem H

(1) Integrality. b_μ is admissible, so $E_k b_\mu$ is admissible (Prop. 3.2) and by Lemma 2.2 every b -coordinate has $v \geq 0$. By Day 214 §8.1 (Gauss's lemma over $\mathbb{Q}[s, t]$), $E_k e_\mu \in \bigoplus \mathbb{Q}[s, t] e_\nu$, so the b_ν -coordinate of $E_k b_\mu$ is $p(s, t)/s^{n(\nu)-n(\mu)}$ with $p \in \mathbb{Q}[s, t]$. Its s -adic valuation over $\mathbb{Q}(t)$ is ≥ 0 , so the s^j -coefficients of p for $j < n(\nu) - n(\mu)$ vanish in $\mathbb{Q}(t)$, hence in $\mathbb{Q}[t]$; the coordinate lies in $\mathbb{Q}[s, t]$. As $\mathcal{L}$ is closed under $\mathcal{O}$ -combinations, $E_k \mathcal{L} \subseteq \mathcal{L}$.

(3) Explicit matrix. By Lemma 2.2 the $\bar{b}_\nu$ -coordinate of $L_k \bar{b}_\mu$ is in $(E_k b_\mu)(\nu')$. Apply (2.3) with $\text{in}(b_\mu)(\rho) = [\rho = \mu']$: the result is $t^{\text{inv}(\nu'/\mu')} \prod \left[\begin{smallmatrix} m_v(\nu') \\ r_v \end{smallmatrix} \right]_t$ if ν'/μ' is a vertical k -strip, and 0 otherwise.

(2) The intertwining.

1. *The inv count.* Let κ/ρ be a vertical k -strip with data (r_v) and A as above. Using $n(\kappa) = \sum_{\text{pairs}} \min$, lowering the parts in A by one changes a pair's min by: -1 if both parts are in A ; -1 iff $v \leq w$ if part $i \in A$ has value v and part $j \notin A$ has value w ; 0 otherwise. Hence

$$n(\kappa) - n(\rho) = \binom{k}{2} + \text{inv}(\kappa/\rho) + \sum_v r_v(m_v - r_v). \quad (3.1)$$

2. *Flip t .* Since $\left[\begin{smallmatrix} m \\ r \end{smallmatrix} \right]_{t^{-1}} = t^{-r(m-r)} \left[\begin{smallmatrix} m \\ r \end{smallmatrix} \right]_t$,

$$t^{\text{inv}} \prod_v \left[\begin{smallmatrix} m_v \\ r_v \end{smallmatrix} \right]_t = t^{n(\kappa) - n(\rho) - \binom{k}{2}} \psi_{\kappa/\rho}(t^{-1}), \quad \psi_{\kappa/\rho}(u) := \prod_v \left[\begin{smallmatrix} m_v(\kappa) \\ r_v \end{smallmatrix} \right]_u.$$

3. *Pieri.* Macdonald III (3.2): $e_k P_\rho(x; u) = \sum_{\kappa/\rho \text{ vert. } k\text{-strip}} \psi_{\kappa/\rho}(u) P_\kappa(x; u)$, with $m_v(\kappa) = \kappa'_v - \kappa'_{v+1}$ and $r_v = \kappa'_v - \rho'_v$ (machine-checked, §6).
4. *Conclude.* For $X \in \mathcal{L}$ write $h(\kappa) := t^{-n(\kappa)} \text{in}(X)(\kappa)$, so $\varphi(\bar{X}) = \sum_\kappa h(\kappa) P_\kappa(x; t^{-1})$ by Lemma 2.2. By (2.3) and step 2,

$$t^{-n(\kappa)} \text{in}(E_k X)(\kappa) = t^{-\binom{k}{2}} \sum_\rho \psi_{\kappa/\rho}(t^{-1}) h(\rho),$$

$$\text{so } \varphi(L_k \bar{X}) = t^{-\binom{k}{2}} \sum_\rho h(\rho) \sum_\kappa \psi_{\kappa/\rho}(t^{-1}) P_\kappa(x; t^{-1}) = t^{-\binom{k}{2}} e_k \varphi(\bar{X}). \quad \square$$

5 Proof of the Corollary

1. *Iterate.* $e_\lambda^* = E_{\lambda_1} \cdots E_{\lambda_\ell}(1)$ with $1 = b_\emptyset$ and $\varphi(\bar{b}_\emptyset) = 1$. Iterating (H), $\varphi(e_\lambda^* \bmod s) = t^{-\sum_i \binom{\lambda_i}{2}} e_\lambda = t^{-n(\lambda')} e_\lambda$.
2. *Read off d .* By Day 214 Theorem 2(a), $e_\lambda^* = \sum_\mu s^{n(\mu)} d_{\lambda\mu}(t)(1 + O(s))e_\mu$, so its class mod s is $\sum_\mu d_{\lambda\mu} \bar{b}_\mu$. Comparing $P_{\mu'}(x; t^{-1})$ -coefficients, $t^{-n(\mu')} d_{\lambda\mu} = t^{-n(\lambda')} [P_{\mu'}(x; t^{-1})] e_\lambda$.
3. *Kostka–Foulkes.* With $e_\lambda = \sum_\nu K_{\nu'\lambda} s_\nu$ and $s_\nu = \sum_\kappa K_{\nu\kappa}(u) P_\kappa(x; u)$ (Macdonald III (2.6)), $u = t^{-1}$:

$$d_{\lambda\mu} = t^{-n(\lambda')} \sum_\nu K_{\nu'\lambda} t^{n(\mu')} K_{\nu\mu'}(t^{-1}) = t^{-n(\lambda')} \sum_\nu K_{\nu'\lambda} \tilde{K}_{\nu\mu'}(t).$$

4. *Consequences.* $\tilde{K} \in \mathbb{N}[t]$ (Lascoux–Schützenberger cocharge, Macdonald III (6.5)) and $d \in \mathbb{Q}[t]$ (Day 214) give $d \in \mathbb{N}[t]$. At $t = 1$: $\sum_\nu K_{\nu'\lambda} K_{\nu\mu'} = M_{\lambda\mu'}$ by dual RSK / Macdonald I (6.6)–(6.7), via $e_\lambda = \sum M_{\lambda\kappa} m_\kappa$ and $P_\kappa(x; 1) = m_\kappa$. $\square$

6 Verification (scripts/day215b/, scripts/day215/)

- `op_kill.py`: explicit matrix (3) against the exact engine (0.1) applied to b_μ ; b -coordinates checked exactly with no negative s -power (the integrality assert), and the $s = 0$ value compared with the prediction.
- `hl_pieri_check.py`: Macdonald III (3.2) with P built from the (2.2) symmetrisation.

Check log (stated honestly).

- `op_kill_sym5.log` (symbolic s, t , all $n + k \leq 5$): **26/26 pass, 0 bad**. Complete; this is the load-bearing check.
- `op_kill_num6.log` (numeric t , $n + k \leq 6$): **PARTIAL** — through $n = 0..3$ (28 cases, 0 bad); no lines for $n = 4, 5$.
- `op_kill_num7.log` (numeric t , $n + k \leq 7$): **PARTIAL** — through $n = 0, 1$ (13 cases, 0 bad).
- `hl_pieri_num6.log`: Macdonald III (3.2), $n = 1..6$, 0 bad (complete); symbolic t , $n \leq 4$: 0 failures.
- Independent end-to-end check of the Corollary (including the normalisation $t^{-n(\lambda')}$): **233/233** pairs with $n \leq 7$ (`scripts/day215/test_formula.py`).

7 Scope, grade, and the novelty question

Grade: proved (self-proved; sober re-read once; not peer-reviewed). No known logical gap. External inputs: 207b (proved, Clio-reviewed); Day 214 §§1, 5, 8.1–8.3 (proved, review queued); textbook Macdonald III (2.2), (2.6), (3.2), (6.5) and I (1.11), (6.6); (3.2) is load-bearing and machine-checked.

Novelty: UNAUDITED. This may well be folklore: it looks like a product-level shadow of $P_\lambda(x; q = 0, t) = \text{Hall–Littlewood}$. Candidates to check: Hikita 2503.23597 (Lemma 6.3 covers only the $q \rightarrow \infty$ limit on the unrescaled e); Bechtloff Weising–Orr 2410.13642; Shimozono–Zabrocki; general “ $q \rightarrow 0/q \rightarrow \infty$ of Macdonald operators gives HL” (Macdonald VI). The features that look specific here are the rescaling $b_\mu = s^{n(\mu)}e_\mu$ and the conjugation $\mu \mapsto \mu'$.

Remark (Mechanism). The rescaling $b_\mu = s^{n(\mu)}e_\mu$ is exactly the Gauss-valuation weight $\omega(\beta) = n(\text{sort}(\beta)')$. The tilt $c = \alpha - \frac{1}{2}$ turns each kernel factor into 1, t , or the level-set HL kernel, so the $s \rightarrow 0$ limit sees only the relative order of exponents — precisely the information HL symmetrisation uses.